5.2 The impacts of climate change at a range of geographic scales

This chapter will explain the present-day and predicted future impacts of climate change:

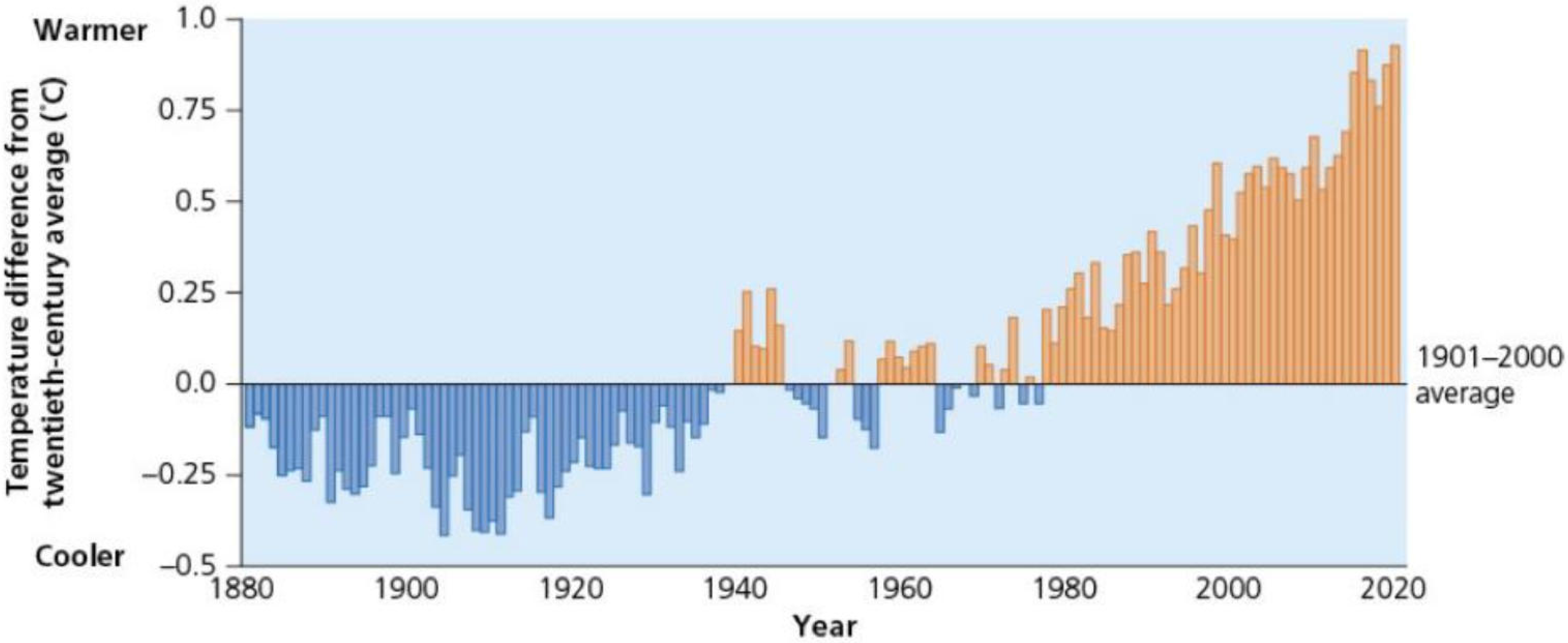
★ the changes to global temperature and weather patterns
★ the changes to sea levels
★ future predictions.

The changes to global temperature and weather patterns

Figure 5.9 shows the change in global surface temperatures between 1880 and 2020. Between 1880 and 1940, annual temperatures were below the 1880–2020 long-term average. It was coldest (about 0.4°C colder) between about 1900 and 1910. Temperatures generally fluctuated above and below the long-term average between 1940 and 1980. However, since 1980 temperatures have always been higher than the long-term average, reaching approximately 1.0°C hotter than the long-term average around 2015. It is predicted to be about 1.1–1.2°C above the long-term average in 2025.

According to a 2021 report by the IPCC (the United Nations body that assesses the science for climate change), the enhanced greenhouse effect should be seen as a 'code red' for humanity. It warns of increasingly extreme heatwaves, droughts, flooding and high temperatures over the next few decades. It also warns of an impending **tipping point**. It points out that:

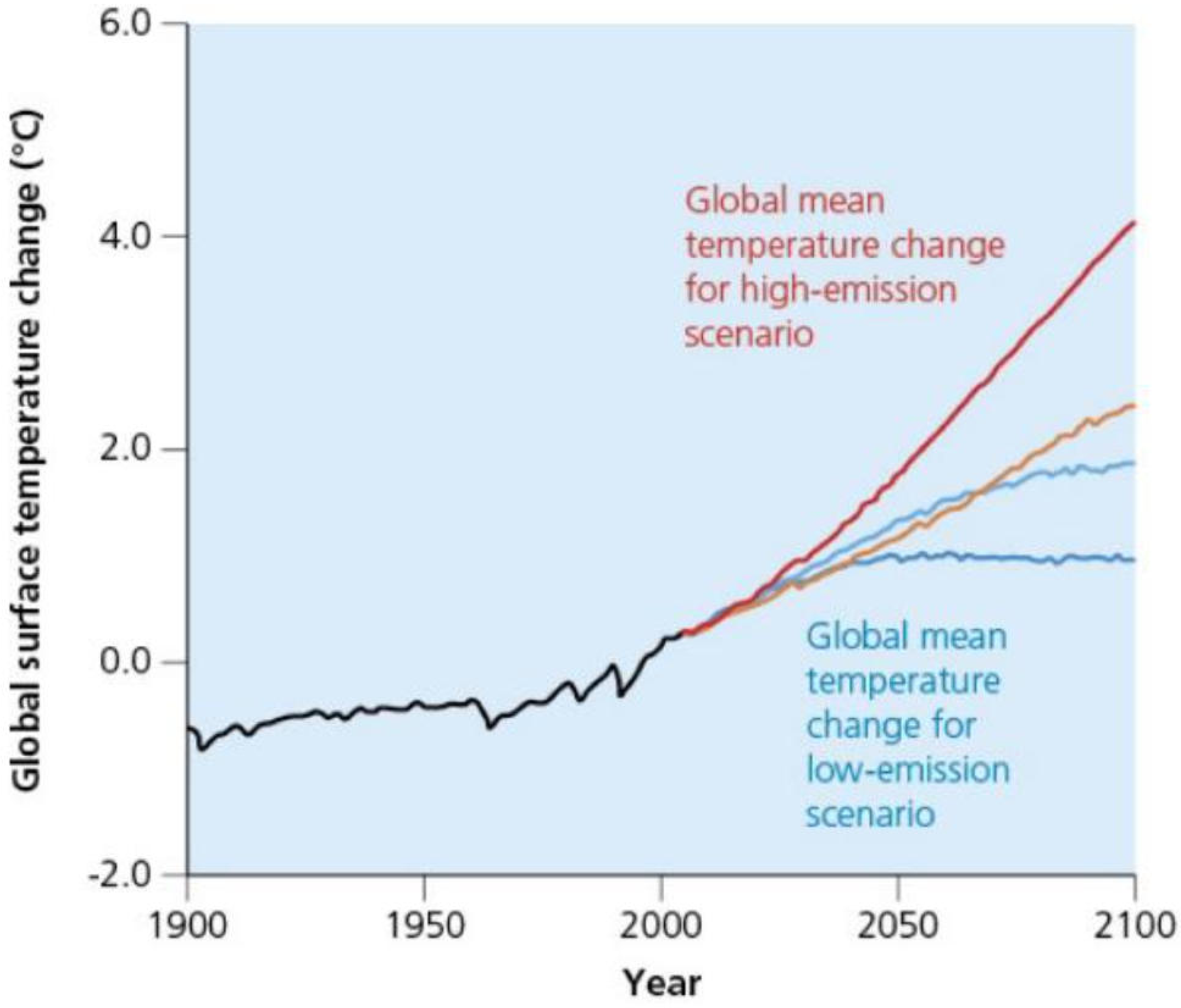
» since 1970, global surface temperatures have risen faster than in any other 50-year period over the last 2000 years
» global surface temperatures are about 1.1–1.2°C higher than in 1850–90
» the years 2017–24 were the warmest on record since 1850
» the rate of sea level rise has increased more than three-fold since 1971 – over 260 million people are at risk from rising sea levels and this could rise to 400 million people by 2100
» human influence is more than 90 per cent likely to be the cause of the global retreat of glaciers and the decrease of Arctic sea ice
» hot extremes have become more frequent and intense since the 1950s, for example there were droughts in the Amazon in 2005, 2010, 2015 and 2020, whereas cold extremes have become less frequent and less intense.



▲ **Figure 5.9** Annual global surface temperature change for land and ocean

The report also states that the enhanced greenhouse effect is much worse than previously thought, and that 40 per cent of the world's population are vulnerable to climate change. The report claims that climate extremes are going beyond humanity's and species' ability to cope. Between 2000 and 2020 there was a fifteen-fold increase in the number of deaths due to storms, floods and droughts, especially in emerging countries.

Extremes of weather are becoming more frequent and severe. The heatwave that killed over 70,000 people in Europe in 2003 was said to be a one-in-a-thousand-year event – but now it is said to be a one-in-ten-year event. In 2024, around 1300 pilgrims who made the journey to Mecca died due to heat exhaustion and dehydration. Temperatures there reached 50°C in the shade.

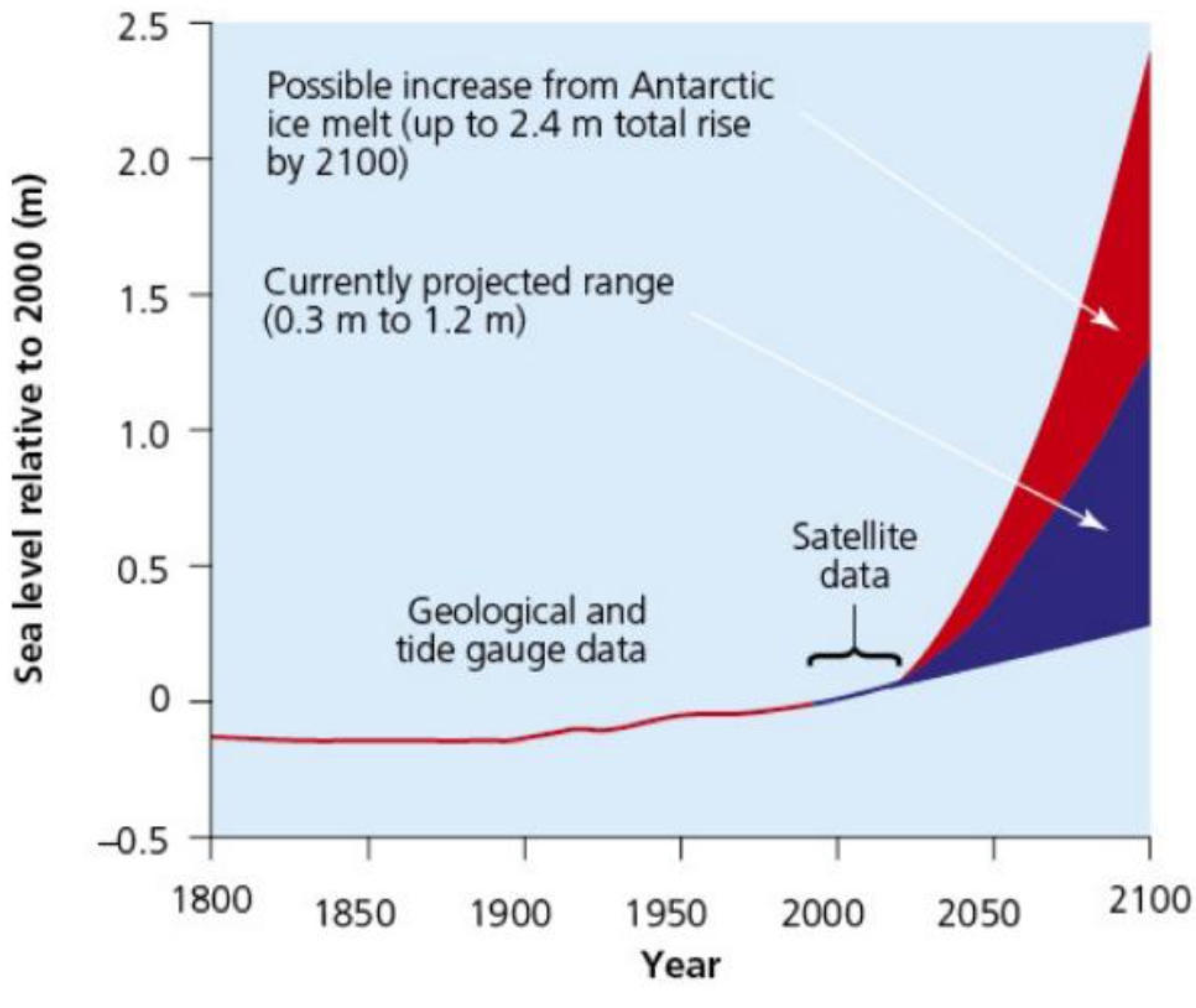


▲ **Figure 5.10** Predicted temperature changes 2000–2100 under high- and low-emissions scenarios

The changes to sea levels

Global mean sea level has risen about 21–24 cm since 1880. As already stated, the rising water level is mostly due to a combination of meltwater from glaciers and ice sheets and thermal expansion of sea water as it warms (the steric effect). Another, much smaller, contributor to sea level rise is the decline in the amount of water on land – in aquifers, lakes and reservoirs, rivers and soil moisture. This shift of liquid water from land to ocean is largely due to people depleting groundwater.

The global mean sea level in the ocean rose by 3.6 mm per year from 2006 to 2015, which was 2.5 times the average rate of 1.4 mm per year throughout most of the twentieth century. By 2100, global mean sea level is likely to rise by at least 0.3 m above 2000 levels (Figure 5.11). In the United States, about 30 per cent of the population lives in areas where sea level plays a role in flooding, shoreline erosion and hazards from storms.



▲ **Figure 5.11** Global mean sea level rise, 1800–2100 (projected)

Over 260 million people are at risk from rising sea levels, and by the year 2100 this could rise to over 400 million people. Over 60 per cent of these live in tropical regions. Many Pacific Ocean and Indian Ocean low-lying islands and many of the world's megacities are less than 10 m above sea level and coastal flooding is a major risk. In addition, food production in coastal areas is likely to be affected by saltwater intrusion. Extreme events that previously occurred once a century on average, are now predicted to occur annually. Up to 1 billion people are at risk from coastal-specific climate hazards such as tropical cyclones and storm surges.

Future predictions

Greenhouse gas concentrations in the atmosphere will continue to increase unless annual emissions decrease substantially. Increased concentrations are expected to raise average temperatures, alter patterns and amounts of precipitation, reduce ice and snow cover, increase the frequency, intensity and/or duration of extreme events, increase threats to human health, increase the acidity of oceans and cause ecosystems to shift their locations.

Precipitation is likely to increase, especially in tropical and high-latitude regions. The strength of the winds associated with tropical storms is likely to increase, although some scientists believe the storms will be more frequent but not necessarily stronger.

Sea ice and snow cover in the Northern Hemisphere has decreased since about 1970. Over the next century, these trends will continue.

Food production

Food production is likely to be affected by rising global temperatures. The decline in water resources may make it difficult for many farmers to continue the type of farming they currently practise. They may have to change the type of farming or the crops they grow, or they may even be forced out of farming altogether. Places in the Middle East, such as in Egypt, Dubai and Saudi Arabia, are likely to experience extreme heatwaves and farmlands may become desertified. In parts of Africa, especially Southern Africa, cereal crop productivity will be reduced as 30 per cent of the area that is used for maize production and 50 per cent of the area that is used for beans is likely to go out of production due to the enhanced greenhouse effect.

Activities

1 Compare the greenhouse effect and the enhanced greenhouse effect.

2 Using Figure 5.11 (page 116), explain why there are variations in the predicted sea level rise by 2100.